

A Practical Guide to Sentiment analysis

TOPIC -- SENTIMENT ANALYSIS

-- 01 --

Ethical considerations Issues such as privacy, consent, and bias are crucial since sentiment analysis regularly analyzes personal data without explicit user consent. The potential for misinterpretation and misuse of sentiment data can significantly impact societal norms. Furthermore, the development of ethical frameworks, as seen in projects like SEWA, where Ethical and Industrial Valorisation Advisory Boards are established, is essential for addressing these challenges. These boards help ensure that sentiment analysis technologies are used responsibly, especially in applications involving the recognition of human emotions and behaviors. Such frameworks are vital for guiding the responsible use of sentiment analysis tools, ensuring they promote equity and respect user autonomy, and effectively address both routine and complex ethical issues.

-- 02 --

Meta-Bootstrapping by Riloff and Jones in 1999. Level One: Generate extraction patterns based on the pre-defined rules and the extracted patterns by the number of seed words each pattern holds. Level Two: Top 5 words will be marked and add to the dictionary. Repeat. Basilisk (Bootstrapping Approach to Semantic Lexicon Induction using Semantic Knowledge) by Thelen and Riloff. Step One: Generate extraction patterns. Step Two: Move best patterns from Pattern Pool to Candidate Word Pool. Step Three: Top 10 words will be marked and add to the dictionary. Repeat. Overall, these algorithms highlight the need for automatic pattern recognition and extraction in subjective and objective task. Subjective and object classifier can enhance the several applications of natural language processing. One of the classifier's primary benefits is that it popularized the practice of data-driven decision-making processes in various industries. According to Liu, the applications of subjective and objective identification have been implemented in business, advertising, sports, and social science.

-- 03 --

Online review classification: In the business industry, the classifier helps the company better understand the feedbacks on product and reasonings behind the reviews. Stock price prediction: In the finance industry, the classifier aids the prediction model by process auxiliary information from social media and other

textual information from the Internet. Previous studies on Japanese stock price conducted by Dong et al. indicates that model with subjective and objective module may perform better than those without this part. Social media analysis. Students' feedback classification. Document summarising: The classifier can extract target-specified comments and gathering opinions made by one particular entity. Complex question answering. The classifier can dissect the complex questions by classing the language subject or objective and focused target. In the research Yu et al.(2003), the researcher developed a sentence and document level clustered that identity opinion pieces. Domain-specific applications. Email analysis: The subjective and objective classifier detects spam by tracing language patterns with target words.

-- 04 --

Feature/aspect-based It refers to determining the opinions or sentiments expressed on different features or aspects of entities, e.g., of a cell phone, a digital camera, or a bank. A feature or aspect is an attribute or component of an entity, e.g., the screen of a cell phone, the service for a restaurant, or the picture quality of a camera. The advantage of feature-based sentiment analysis is the possibility to capture nuances about objects of interest. Different features can generate different sentiment responses, for example a hotel can have a convenient location, but mediocre food. This problem involves several sub-problems, e.g., identifying relevant entities, extracting their features/aspects, and determining whether an opinion expressed on each feature/aspect is positive, negative or neutral. The automatic identification of features can be performed with syntactic methods, with topic modeling, or with deep learning. More detailed discussions about this level of sentiment analysis can be found in Liu's work.

-- 05 --

Example of an objective sentence: 'To be elected president of the United States, a candidate must be at least thirty-five years of age.' The term subjective describes the incident contains non-factual information in various forms, such as personal opinions, judgment, and predictions, also known as 'private states'. In the example down below, it reflects a private states 'We Americans'. Moreover, the target entity commented by the opinions can take several forms from tangible product to intangible topic matters stated in Liu (2010). Furthermore, three types of attitudes were observed by Liu (2010), 1) positive opinions, 2) neutral opinions, and 3) negative opinions.

-- 06 --

Metaphorical expressions. The text contains metaphoric expression may impact on the performance on the extraction. Besides, metaphors take in different forms, which may have been contributed to the increase in detection. Discrepancies in writings. For the text obtained from the Internet, the discrepancies in the writing style of targeted text data involve distinct writing genres and styles. Context-sensitive. Classification may vary based on the subjectiveness or objectiveness of previous and following sentences. Time-sensitive attribute. The task is challenged by some textual data's time-sensitive attribute. If a group of researchers wants to confirm a piece of fact in the news, they need a longer time for cross-validation, than the news becomes outdated. Cue words with fewer usages. Ever-growing volume. The task is also challenged by the sheer volume of textual data. The textual data's ever-growing nature makes the task overwhelmingly difficult for the researchers to complete the task on time. Previously, the research mainly focused on document level classification. However, classifying a document level suffers less accuracy, as an article may have diverse types of expressions involved. Researching evidence suggests a set of news articles that are expected to dominate by the objective expression, whereas the results show that it consisted of over 40% of subjective expression. To overcome those challenges, researchers conclude that classifier efficacy depends on the precisions of patterns learner. And the learner feeds with large volumes of annotated training data outperformed those trained on less comprehensive subjective features. However, one of the main obstacles to executing this type of work is to generate a big dataset of annotated sentences manually. The manual annotation method has been less favored than automatic learning for three reasons:

-- 07 --

Sentiment analysis (also known as opinion mining) is the use of natural language processing, text analysis, computational linguistics, and biometrics to systematically identify, extract, quantify, and study affective states and subjective information. Sentiment analysis is widely applied to voice of the customer materials such as reviews and survey responses, online and social media, and healthcare materials for applications that range from marketing to customer service to clinical medicine. With the rise of deep language models, such as RoBERTa, more difficult data domains can be analyzed, e.g., news texts where authors typically express their opinion/sentiment less explicitly.